

A local overlap that passes the test

OHRC to TMC-2 | Frozen middle-region audit | Relative image registration

REGISTERED - LOCAL

The tested middle ROI passes camera/terrain, SIFT, AKAZE, coverage and reverse checks. The full OHRC strip and absolute 5 m lunar position remain ABSTAIN.

1 / What was tested

Real OHRC science pixels were projected through their camera model and a TMC-2 DTM onto the native 5 m TMC-2 ortho grid. The audited OHRC native range is samples 1,000-11,000 and lines 42,000-58,000. The XML footprint intersection is 80.356 km²; that polygon alone is not proof of registration.

Instrument	Exact product ID	Role
OHRC	ch2_ohr_nrp_20241117T2033118394_d_img_d18	Fine-scale optical source
TMC-2	ch2_tmc_ndn_20231101T0125121377_d_oth_d18	5 m ortho target
TMC-2 DTM	ch2_tmc_ndn_20231101T0125121377_d_dtm_d18	Terrain projection

2 / Frozen image evidence

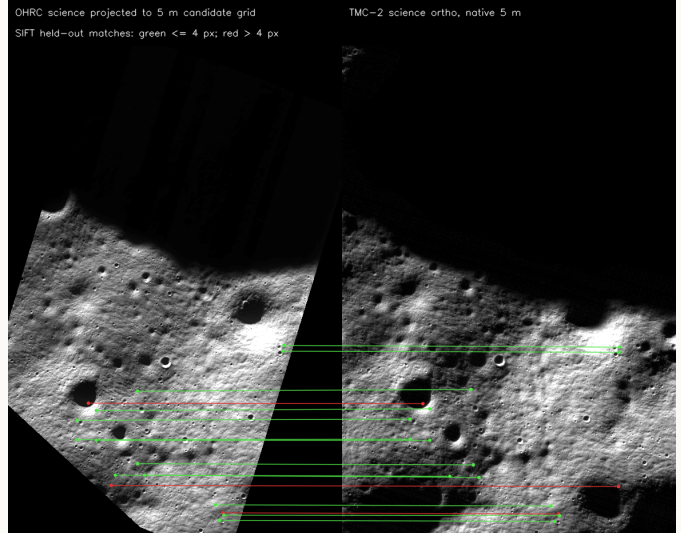
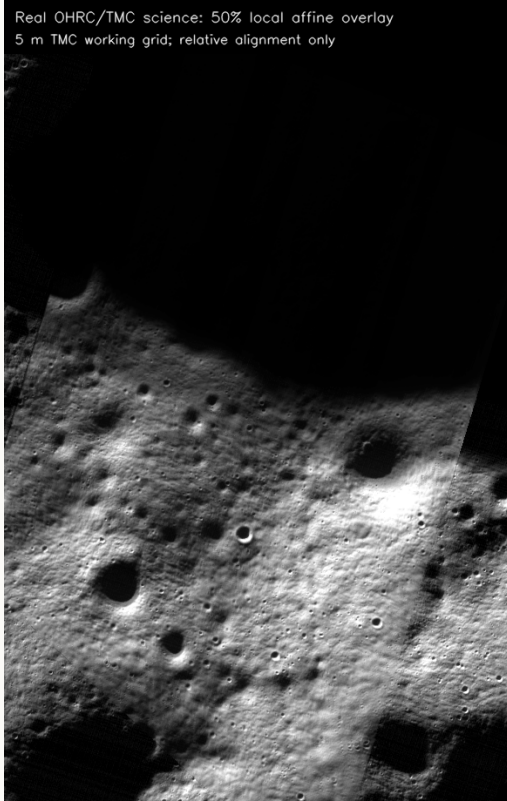
Each method used a fixed reciprocal/ratio search, candidate window and affine RANSAC gate. Held-out spatial bins were excluded from fitting. Errors are in **native 5 m TMC-2 pixels**, not OHRC pixels or absolute lunar ground error.

Method	Train	Held-out	RMSE / P90	Spread
SIFT	785/789	93/111 (83.8%)	1.30 / 1.99 px	11/18; 32.7% hull
AKAZE	362/370	48/56 (85.7%)	1.44 / 2.19 px	10/18; 27.9% hull

Fixed minimum: 6 training inliers at 40%, 3 held-out at 40%, 6/18 occupied cells, 15% hull and 3 TMC-pixel median reverse closure. Separately fitted reverse medians: SIFT 1.26 px; AKAZE 2.20 px. The 878 and 410 accepted match counts overlap and cannot be added as distinct ground-control points.

Terrain agreement at the tested scale

The saved figures below come from the frozen science-pixel audit. Image alignment and match support are shown separately so a visual overlay cannot substitute for the fixed tests.



Left: saved local affine overlay. Right: held-out SIFT correspondences. Both are science-data figures, not a full-strip accuracy map.

3 / Independent checks and evidence boundary

The DTM navigation was compared with LOLA terrain, and OHRC was screened independently against LOLA hillshade at 120 m cells. Alternative TMC locations 20 km north and 10 km south failed the same matcher settings. These support the local association without establishing surveyed 5 m absolute coordinates.

Scope of the decision: relative local OHRC-to-TMC-2 correspondence only. Prior full-strip affine transfer failed outside this middle ROI. The local affine can absorb camera/terrain residuals. No IIRS science-pixel registration is claimed here.

4 / Reproducibility

Audit ID: **REALPAIR005_OHRC_TMC_ALIGNED_MIDDLE_VERIFICATION_V1**. Fixed gates are in `aligned_candidate_config_v1.json`. SIFT and AKAZE each retain `config.json`, `matches.csv`, `inliers.csv`, `reverse_matches.csv`, `transform.json` and `metrics.json`. The scoped decision is `aligned_middle_decision.json`; `aligned_middle_sha256_v2.txt` binds derived artifacts. The original science labels and product hashes are listed in the full scientific verification document.

Source: `data/derived/REALPAIR005_science_v1/ALIGNED_MIDDLE_SCIENTIFIC_VERIFICATION.md`